

CHANGJIAN LI

School of Informatics, 10 Crichton Street, Newington, Edinburgh, EH8 9AB, United Kingdom

+44(0)1316504450 ◊ Changjian.li@ed.ac.uk ◊ [Homepage](#)

ACADEMIC POSITIONS

Assistant Professor

Oct. 2022 - present

School of Informatics, University of Edinburgh, UK

Starting Researcher

Oct. 2021 - Jul. 2022

GraphDeco Group, INRIA Sophia-Antipolis, France

Work with Dr. Adrien Bousseau

Postdoctoral Researcher

Oct. 2019 - Sep. 2021

Smart Geometry Processing Group, University College London, UK

Hosted by Prof. Niloy Mitra

EDUCATION

The University of Hong Kong, Hong Kong, China

Aug. 2014 - Sep. 2019

Ph.D., Department of Computer Science

Supervisor: Prof. Wenping Wang

Shandong University, Jinan, China

Sep. 2010 - Jun. 2014

BEng, Software Engineering

Software College

PUBLICATION LIST

**corresponding author, #joint first author*

1. Yi Yang, Hao Pan, Yijing Cui, Alla Sheffer, **Changjian Li**. *Sketch2Arti: Sketch-based Articulation Modeling of CAD Objects*. ACM Trans. Graph., 45(4), 2026, proceedings of SIGGRAPH, 2026.
2. Yilin Liu, Niladri Shekhar Dutt, **Changjian Li**, Niloy J. Mitra. *B-repLer: Language-guided Editing of CAD Models*. ACM SIGGRAPH Conference Proceedings, 2026.
3. Haocheng Yuan, Adrien Bousseau, Hao Pan, Lei Zhong, **Changjian Li**. *DancingBox: A Lightweight MoCap System for Character Animation from Physical Proxies*. ACM Conference on Human Factors in Computing Systems (CHI), 2026. **(Best Paper Honorable Mention Award)**.
4. Ankan Bhunia, **Changjian Li**, Hakan Bilen. 2025. *Interactive Anomaly Detection for Articulated Objects via Motion Anticipation*. The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.
5. Jiawei Wang, Zhiming Cui, **Changjian Li**. 2025. *VQ-SGen: A Vector Quantized Stroke Representation for Creative Sketch Generation*. International Conference on Computer Vision (ICCV), 2025.
6. Miaowei Wang, **Changjian Li**, Amir Vaxman. 2025. *CanFields: Consolidating Diffeomorphic Flows for Non-Rigid 4D Interpolation from Arbitrary-Length Sequences*. International Conference on Computer Vision (ICCV), 2025.
7. Yulong Dou, Han Wu, Changjian Li, Chen Wang, Tong Yang, Min Zhu, Dinggang Shen, and Zhiming Cui. *CLIK-Diffusion: Clinical Knowledge-informed Diffusion Model for Tooth-Alignment*. Medical Image Analysis (MedIA), 2025.

8. Salvatore Esposito, Daniel Rebain, Aron Onken, **Changjian Li**, Oisin Mac Aodha. 2025. *VesselSDF: Distance Field Priors for Vascular Network Reconstruction*. International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2025.
9. Lei Zhong, Chuan Guo, Yiming Xie, Jiawei Wang, **Changjian Li**. 2025. *Sketch2Anim: Towards Transferring Sketch Storyboards into 3D Animation*. ACM Trans. Graph., 44(4), 2025, proceedings of SIGGRAPH, 2025.
10. Guangshun Wei, Yuan Feng, Long Ma, Chen Wang, Yuanfeng Zhou, **Changjian Li**. 2025. *PC-Dreamer: Point Cloud Completion Through Multi-view Diffusion Priors*. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
11. Thomas Walker*, Salvatore Esposito*, Daniel Rebain, Amir Vaxman, Arno Onken, **Changjian Li**, Oisin Mac Aodha. 2025. *CrossSDF: 3D Reconstruction of Thin Structures From Cross-Sections*. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
12. Ankan Bhunia, **Changjian Li**, Hakan Bilen. 2025. *Odd-One-Out: Anomaly Detection by Comparing with Neighbors*. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
13. Longbin Ji, Lei Zhong, Pengfei Wei, **Changjian Li**. 2025. *PoseTraj: Pose-Aware Trajectory Control in Video Diffusion*. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
14. Duolikun Danier, Mehmet Aygün, **Changjian Li**, Hakan Bilen, Oisin Mac Aodha. 2025. *DepthCues: Evaluating Monocular Depth Perception in Large Vision Models*. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
15. Haocheng Yuan, Adrien Bousseau, Hao Pan, Quancheng Zhang, Niloy Mitra, **Changjian Li**. 2024. *DiffCSG: Differentiable CSG via Rasterization*. ACM SIGGRAPH Asia Conference Proceedings, 2024.
16. Zhentao Liu, Yu Fang, **Changjian Li**, Han Wu, Yuan Liu, Zhiming Cui, Dinggang Shen. *Geometry-Aware Attenuation Field Learning for Sparse-View CBCT Reconstruction*. IEEE Transactions on Medical Imaging (TMI), 2024.
17. Haocheng Yuan, Jing Xu, Hao Pan, Adrien Bousseau, Niloy Mitra, **Changjian Li**. 2024. *CADTalk: An Algorithm and Benchmark for Semantic Commenting of CAD Programs*. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024. (Highlight, top 10%)
18. Jiawei Wang, **Changjian Li**. 2024. *ContextSeg: Sketch Semantic Segmentation by Querying the Context with Attention*. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
19. Ankan Bhunia, **Changjian Li**, Hakan Bilen. 2024. *Looking 3D: Anomaly Detection with 2D-3D Alignment*. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
20. Guangshun Wei, Hao Pan, Shaojie Zhuang, Yuanfeng Zhou*, **Changjian Li**, 2023. *iPUNet: Iterative Cross Field Guided Point Cloud Upsampling*. IEEE Visualization and Computer Graphics (TVCG), 2023.
21. Felix Hähnlein, **Changjian Li**, Niloy Mitra, Adrien Bousseau. 2022. *CAD2Sketch: Generating Concept Sketches from CAD Sequences*. ACM Trans. Graph., 41(6), 2022, proceedings of SIGGRAPH Asia, 2022.
22. **Changjian Li**, Hao Pan, Adrien Bousseau, Niloy Mitra. 2022. *Free2CAD: Parsing Freehand Drawings into CAD Commands*. ACM Trans. Graph., 41(4), 2022, proceedings of SIGGRAPH, 2022. (Best Paper Honorable Mention Award).
23. Cheng Lin, **Changjian Li***, Yuan Liu, Nenglu Chen, Yi-King Choi, Wenping Wang*. 2021. *Point2Skeleton: Learning Skeletal Representations from Point Clouds*. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2021. (Oral Presentation, Best Paper Nomination Award)

24. Zhiming Cui, **Changjian Li**, Lei Yang, Chunfeng Lian, Feng Shi, Wenping Wang, Dijia Wu, Dinggang Shen. 2021. *VertNet: Accurate Vertebra Localization and Identification Network from CT Images*. International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2021.
25. Zhiming Cui, **Changjian Li**, Zhixu Du, Nenglun Chen, Guodong Wei, Runnan Chen, Lei Yang, Dinggang Shen, Wenping Wang. 2021. *Structure-Driven Unsupervised Domain Adaptation for Cross-Modality Cardiac Segmentation*. IEEE Transactions on Medical Imaging (TMI), 2021.
26. Zhiming Cui[#], Bojun Zhang[#], Chunfeng Lian, **Changjian Li**^{*}, Lei Yang, Min Zhu^{*}, Wenping Wang, Dinggang Shen^{*}. 2021. *Hierarchical Morphology-Guided Tooth Instance Segmentation from CBCT Images*. Information Processing in Medical Imaging (IPMI) 2021. (Oral Presentation)
27. Zhiming Cui, **Changjian Li**, Nenglun Chen, Guodong Wei, Runnan Chen, Yuanfeng Zhou, Dinggang Shen^{*}, Wenping Wang^{*}. 2021. *TSegNet: an Efficient and Accurate Tooth Segmentation Network on 3D Dental Model*. Medical Image Analysis (MedIA), 2021.
28. Yushiang Wong, **Changjian Li**, Matthias Niessner, Niloy Mitra. 2021. *RigidFusion: RGB-D Scene Reconstruction with Rigidly-moving Objects*. Eurographics (EG), 2021.
29. Hui Zhang, Lei Yang, **Changjian Li**, Bojian Wu, Wenping Wang. 2021. *ScaffoldGAN: Synthesis of Scaffold Materials based on Generative Adversarial Networks*. Computer-Aided Design (CAD), 2021.
30. **Changjian Li**, Hao Pan, Adrien Bousseau, Niloy Mitra. 2020. *Sketch2CAD: Sequential CAD Modeling by Sketching in Context*. ACM Trans. Graph., 39(6), 2020, proceedings of SIGGRAPH Asia, 2020.
31. Cheng Lin, Lingjie Liu, **Changjian Li**, Leif Kobbelt, Bin Wang, Shiqing Xin, Wenping Wang. 2020. *SEG-MAT: 3D Shape Segmentation Using Medial Axis Transform*. IEEE Visualization and Computer Graphics (TVCG), 2020).
32. Zhiming Cui, **Changjian Li**, Wenping Wang. 2019. *ToothNet: Automatic Tooth Instance Segmentation and Identification from Cone Beam CT Images*. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2019.
33. Cheng Lin, **Changjian Li**, Wenping Wang. 2019. *Floorplan-Jigsaw: Jointly Estimating Scene Layout and Aligning Partial Scans*. International Conference on Computer Vision (ICCV), 2019.
34. **Changjian Li**, Hao Pan, Yang Liu, Xin Tong, Alla Sheffer, Wenping Wang. 2018. *Robust Flow-Guided Neural Prediction for Sketch-Based Freeform Surface Modeling*. ACM Trans. Graph., 37(6), 2018, proceedings of SIGGRAPH Asia 2018.
35. **Changjian Li**, Hao Pan, Yang Liu, Xin Tong, Alla Sheffer, Wenping Wang. 2017. *BendSketch: Modeling Freeform Surfaces Through 2D Sketching*. ACM Trans. Graph., 36(4), 2017, proceedings of SIGGRAPH 2017.
36. Hao Pan, Yang Liu, Alla Sheffer, Nicholas Vining, **Changjian Li**, Wenping Wang. 2015. *Flow Aligned Surfacing of Curve Networks*. ACM Trans. Graph., 36(4), 2015, proceedings of SIGGRAPH 2015.
- Preprint
37. Anran Qi, Changjian Li, Adrien Bousseau, and Niloy J. Mitra. *Beyond the Visible: Disocclusion-Aware Editing via Proxy Dynamic Graphs*. ArXiv preprint, 2026.
38. Lei Zhong, Yi Yang, **Changjian Li**. *SMooGPT: Stylized Motion Generation using Large Language Models*. ArXiv preprint, 2026.
39. Danier, Duolikun, Ge Gao, Steven McDonagh, Changjian Li, Hakan Bilen, and Oisin Mac Aodha. *View-Consistent Diffusion Representations for 3D-Consistent Video Generation*. ArXiv preprint, 2025.

40. Yu Fang, Lanzhu Mei, **Changjian Li**, Yuan Liu, Wenping Wang, Zhiming Cui, Dinggang Shen. *SNAF: Sparse-view CBCT Reconstruction with Neural Attenuation Fields*. *ArXiv preprint*, 2022.
41. Yuezhi Yang, Zhiming Cui, **Changjian Li**, Wenping Wang. *ToothInpaintor: Tooth Inpainting from Partial 3D Dental Model and 2D Panoramic Image*. *ArXiv preprint*, 2022.

RESEARCH FUNDING

Adobe Gift	2026
SnapChat Gift	2025
Adobe Gift	2025
Seed Grant, PI - £15,000 GBP <i>Generative AI Laboratory (GAIL), University of Edinburgh</i>	2024
Seed Grant, PI - £6,000 GBP <i>Bayes Center, University of Edinburgh</i>	2023
Start-Up Grant <i>University of Edinburgh</i>	2022

AWARDS AND PRIZES

Best Paper Honorable Mention - DancingBox, ACM SIGCHI	2026
Young Investigator Award - Solid Modeling Association (SMA)	2025
Best Paper Honorable Mention - Free2CAD, ACM SIGGRAPH	2022
Best Paper Candidates - Point2Skeleton, CVPR	2021
Back cover image on Proceedings of ACM SIGGRAPH	2018
YS & Christabel Lung PG Scholarship (HKU)	2014-2017
Postgraduate Scholarship (HKU)	2014-2019
President's Scholarship of Shandong University (top 0.5% , highest student honor)	2013
Google Excellent Student Scholarship (top 0.5%)	2013
Nation Scholarship of China (top 2% , twice)	2011, 2012
Excellent Student of Shandong Province (top 0.05%)	2012

RESEARCH EXPERIENCE

Research Intern <i>Microsoft Research Asia, Beijing</i>	Feb. 2016 - Aug. 2018 <i>Multiple Times</i>
- Mentor: Dr. Hao Pan, Dr. Yang Liu, and Dr. Xin Tong	
Research Assistant <i>The University of Hong Kong, Hong Kong</i>	Feb. 2014 - Jun. 2014
- Supervisor: Prof. Wenping Wang	
- Bachelor Thesis: Sketch-based Modeling: An Intuitive Modeling Interface.	

INVITED TALKS

GAMES: Graphics And Mixed Environment Seminar, China Sketch2Anim: Towards Transferring Sketch Storyboards into 3D Animation.	Jul. 2025
GAMES: Graphics And Mixed Environment Seminar, China CADTalk: An Algorithm and Benchmark for Semantic Commenting of CAD Programs.	Jun. 2024
Keynote Speaker at DIRA Workshop, ECCV2022, Tel Aviv High-quality CAD modeling via Rough Sketching.	Oct. 2022
Invited Workshop Keynote Speaker at GDC, Qingdao, China Free2CAD: Parsing Freehand Drawings into CAD Commands.	Aug. 2022
School of Software, Tsinghua University, China Creating, Analyzing and Processing 3D Data: Applications to interactive 3D modeling and to medical imaging.	Dec. 2021
School of Informatics, University of Edinburgh, UK Creating, Analyzing and Processing 3D Data: Applications to interactive 3D modeling and to medical imaging.	Oct. 2021
Stanford Human-Computer Interaction Group, Stanford, US Sketch2CAD: Sequential CAD Modeling by Sketching in Context.	Jan. 2021
AI Lab, Autodesk, US Sketch2CAD: Sequential CAD Modeling by Sketching in Context.	Jan. 2021
Software College, Shandong University, China Sketch-based Freeform Surface Modeling.	Oct. 2019
iDDA, The Chinese University of Hong Kong (Shenzhen), China SketchAShape: Sketch-based Freeform Surface Modeling.	Jul. 2019
IRC Lab, Shandong University, China Robust Flow-Guided Neural Prediction for Sketch-Based Freeform Surface Modeling. Automatic Tooth Instance Segmentation and Identification from CBCT Images.	Mar. 2019
DGP Lab, University of Toronto, CA BendSketch: Modeling Freeform Surfaces Through 2D Sketching.	Sept. 2018
GAMES: Graphics And Mixed Environment Seminar, China Robust Flow-Guided Neural Prediction for Sketch-Based Freeform Surface Modeling.	Nov. 2018

IN THE PRESS

Google PhD Fellow 2025. Award Recipient - Lei Zhong (PhD student) , and the Informatics News .	2025
ACM SIGGRAPH Technical Paper Press Releases. 3D Generative AI Transforms How We Create, Design, Interact With Digital Content (Sketch2Anim) .	2025
ACM SIGGRAPH Technical Papers Awards. Best Paper Honorable Mention Award - Free2CAD .	2022

CCTV News Report. World Artificial Intelligence Conference (WAIC) - SketchCNN.	2019
Computer Vision News, The magazine of the algorithm community. ICCV Daily 2019. - Floorplan-Jigsaw.	2019
IEEE Computer Society. Poster Sessions Provoke Deep Discussions at the 2019 Conference on CVPR - ToothNet.	2019
ACM SIGGRAPH Asia Technical Paper Press Releases. SketchCNN: Making it Easier To Transform FreeForm 2D Sketching Into 3D Models.	2018
Seamless, Virtual Reality. A new method using CNN to model 3D surface from handwritten 2D sketch.	2018

TEACHING

Lecturer, INFR11132 - Machine Learning Practical. UoE, Informatics.	2025-2026, FY
Lecturer, INFR11212 - Computer Vision. UoE, Informatics.	2024-2025, S2
Lecturer, INFR09051 - Informatics Large Practical. UoE, Informatics.	2024-2025, S1
Lecturer, INFR11140 - Image and Vision Computing. UoE, Informatics.	2023-2024, S2
Lecturer, INFR09051 - Informatics Large Practical. UoE, Informatics.	2023-2024, S1
Assistant, COMP0119 - Acquisition and Processig of 3D Geometry. UCL, CS.	2020-2021, S2
Assistant, COMP2396 - Object Oriented Programming and Java. HKU, CS.	2017-2018, S2
Assistant, ENGG1111 - Computer Programming and Applications. HKU, CS.	2016-2017, S1
Assistant, COMP7507 - Visualization and Visual Analytics. HKU, CS.	2015-2016, S1
Assistant, ENGG1111 - Computer Programming and Applications. HKU, CS.	2014-2015, S1

POSTDOC ADVISING

Duolikun Danier (Co-sup. with Oisin and Hakan)	2024-2026
Guangshun Wei (HKU, remote)	2023-2024

PHD ADVISING

Haocheng Yuan	2023- <i>present</i> , UoE
Ankan Bhunia (co-sup. with Hakan Belien)	2023- <i>present</i> , UoE
Lei Zhong (Google PhD Fellow 2025)	2024- <i>present</i> , UoE
Yi Yang	2026- <i>present</i> , UoE
Yue Ruan (co-sup. with Kartic Surb)	2023-2025, UoE
Miaowei Wang (co-sup. with Amir Vaxman)	2023-2024, UoE
Salvatore Esposito (co-sup. with Arno, Oisin)	2023-2025, UoE
Guangshun Wei (co-sup. with Yuanfeng Zhou)	2021-2023, SDU
Felix Hähnlein (co-sup. with Adrien Bousseau)	2021-2022, INRIA
Yushiang Wong (co-sup. with Niloy Mitra)	2019-2021, UCL
Zhiming Cui (co-sup. with Wenping Wang)	2018-2022, HKU
Cheng Lin (co-sup. with Wenping Wang)	2017-2021, HKU

MASTER ADVISING

Kuankuan Cheng (MScR)	2025.03- <i>present</i> , UoE
Mingjun Yang (MScR)	2023.09-2024.09, UoE
Longbin Ji (MSc)	2024.04-2024.11, UoE
Prakash Nair (MSc)	2024.04-2024.09, UoE

UNDERGRADUATE ADVISING

Jiawei Wang	2023.04-2025.11, SDU
Yi Yang	2024.09-2025.05, UoE
Robin Todd	2024.09-2025.04, UoE
Aulia Ahmad Rafi	2024.09-2025.04, UoE
Jing Xu	2023.10-2023.12, UoE
Shuyuan Zhang	2023.05-2024.05, UoE
Kuankuan Cheng	2023.03-2025.02, UoE
Noah Ramsden	2023.09-2024.04, UoE
Blair Knyspel	2023.09-2024.04, UoE

THESIS COMMITTEE

Fangqiang Ding	May 13, 2025
University of Edinburgh, Ph.D.	
<i>Thesis: Robust Spatial Perception with 4D Radar for Autonomous Driving</i>	
Gen Li	May 5, 2025
University of Edinburgh, Ph.D.	
<i>Thesis: Efficient Affordance Learning from Human-Object Interaction</i>	
Xiang Xu	Apr. 26, 2024
Simon Fraser University, Ph.D.	
<i>Thesis: Towards Automating 3D Computer-Aided Design with Deep Generative Model</i>	
Weiwei Cui	Mar. 15, 2024
Queen Mary, University of London, Ph.D.	
<i>Thesis: AI-based Dentistry Image Segmentation</i>	
Linxu Fan	Jan. 10, 2024
University of Edinburgh, Ph.D.	
<i>Thesis: Simulating Fracture in Brittle Objects and Thin Shell Objects for Visual Effects</i>	
Arushi Goel	Oct. 13, 2023
University of Edinburgh, Ph.D.	
<i>Thesis: Investigating the Role of Linguistic Knowledge in Vision and Language Tasks</i>	
Kiyoong Kim	Oct. 10, 2023
University of Edinburgh, Ph.D.	
<i>Thesis: Towards Accessible Action Recognition</i>	
Zhen Luo	Aug. 30, 2023
University of Edinburgh, MScR	
<i>Thesis: Learning Scene Flow Estimation on mmWave Point Cloud for Robust Human Sensing</i>	
Zhengdi Yu	Jun. 9, 2023
Durham University, MScR	
<i>Thesis: 3D Representation Learning for ShapeReconstruction and Understanding</i>	
Weizhen Ding	May. 28, 2023
The Chinese University of Hong Kong, Shenzhen, MPhil	
<i>Thesis: Colonoscopy Images Analysis via Domain Alignment and Curriculum Learning</i>	

EDITORIAL POSTS, WORKSHOP ORGANIZING

Associate Editor, <i>IEEE TVCG</i>	2024-present
Associate Editor, <i>IEEE CGASI - Intelligent CAD in the Age of AI</i>	2025-2026
Organizer, <i>GenAI for Healthcare Workshop</i>	Jul. 2, 2024

CONFERENCE PROGRAM COMMITTEES

ACM SIGGRAPH: Technical Paper Committee	2026
ACM SIGGRAPH: Poster Jury Committee	2023, 2024
ACM SIGGRAPH Asia: Technical Paper Committee	2023, 2025, 2026
EuroGraphics: International Program Committee	2025
EuroGraphics: State-of-the-art-report IPC	2025
Geometry Design and Computing: Technical Paper Committee	2022, 2023

CONFERENCE SESSION CHAIRING

ACM SIGGRAPH Asia, @Tokyo - Fill the Gap: What Happened In-between?	2024
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REVIEWER SERVICE

ACM Transaction on Graphics (TOG)	2022,2024
ACM SIGGRAPH	2021-2026
ACM SIGGRAPH Asia	2015, 2019-2023, 2025
Conference on Computer Vision and Pattern Recognition (CVPR)	2021,2023,2024,2025,2026
International Conference on Computer Vision (ICCV)	2021, 2025
IEEE Transactions on Visualization & Computer Graphics (TVCG)	2020-2024
Eurographics (EG)	2019-2023, 2025
ACM SIGCHI	2020, 2023
AAAI	2024
ACM Symposium on User Interface Software and Technology (UIST)	2020
Computer-Aided Design	2019, 2022, 2023
Computer Graphics Forum (CGF)	2019
Journal of Computer Science and Technology (JCST)	2019
Pacific Graphics (PG)	2018, 2023, 2024
Eurographics Symposium on Geometry Processing (SGP)	2018, 2023
International Conference on Computer-Aided Design and Graphics (CAD&CG)	2015, 2019
Geometry Design and Computing (GDC)	2022, 2023
IEEE Transactions on Image Processing (TIP)	2023
Journal of Computer Science and Technology (JCST)	2023
The Visual Computer (TVCJ)	2023
International Journal of Computer Vision (IJCV)	2025